

ERMELO, South Africa

WMO: 682670

Lat: 26.4978S

Lon: 29.9839E

Elev: 1774

StdP: 81.75

Time Zone: 2.00 (E02)

Period: 98-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	0.3	1.8	-13.7	1.4	14.0	-11.5	1.7	12.5	12.0	16.0	10.5	14.7	3.6	270	0.517

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	9.5	28.5	14.6	27.3	14.7	26.2	14.8	18.6	23.8	18.1	23.2	17.6	22.5	5.3	290

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
17.1	15.2	20.1	16.6	14.7	19.6	16.1	14.3	19.2	61.0	23.8	59.1	23.4	57.3	22.6	21.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
10.4	9.2	8.2	-2.3	30.7	1.3	1.7	-3.2	31.9	-3.9	32.9	-4.6	33.9	-5.6	35.1
			-4.1	19.6	1.6	0.4	-5.2	19.9	-6.1	20.2	-7.0	20.4	-8.1	20.7
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	15.3	18.6	18.5	17.5	15.3	12.8	10.6	10.1	12.7	15.5	16.6	17.6	18.4
	DBStd	4.08	2.23	1.94	2.20	2.51	2.56	2.48	2.68	3.41	3.86	3.60	3.04	2.32
	HDD10.0	88	0	0	0	2	8	21	33	14	7	3	1	0
	HDD18.3	1253	23	19	39	95	172	233	257	174	96	73	47	26
	CDD10.0	2037	268	237	234	159	94	38	34	99	173	208	229	262
	CDD18.3	159	32	23	14	3	0	0	0	1	12	20	24	29
	CDH23.3	1381	187	140	106	28	6	0	1	23	204	264	212	212
(13)	CDH26.7	167	17	11	7	1	0	0	0	0	23	51	31	28
(14) Wind	WSAvg	4.3	4.1	3.9	3.9	3.8	3.9	4.1	4.2	4.7	5.2	5.0	4.9	4.4
(15) Precipitation	PrecAvg	698	125	77	82	34	10	6	4	10	17	87	105	138
	PrecMax	957	210	238	186	110	53	39	44	51	82	216	202	301
	PrecMin	503	37	17	12	0	0	0	0	0	0	33	33	66
	PrecStd	122	42	45	42	25	12	10	9	14	19	43	43	56
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	28.6	28.1	27.7	26.0	24.0	21.5	21.6	25.3	28.8	30.2	29.6	29.2
		MCWB	16.3	15.9	15.3	14.1	11.5	8.6	7.8	10.7	11.9	13.5	14.9	16.2
	2%	DB	26.8	26.5	25.9	24.2	21.7	19.6	19.6	23.8	27.2	28.2	27.3	27.3
		MCWB	16.6	16.6	15.1	13.4	10.5	8.2	7.7	10.3	11.4	13.4	15.2	16.4
	5%	DB	25.4	25.1	24.4	22.8	20.2	18.1	18.2	22.2	25.9	26.3	25.8	25.8
		MCWB	16.9	16.4	15.1	13.1	9.9	7.6	7.2	9.6	11.3	13.4	15.6	16.6
	10%	DB	24.0	23.7	23.1	21.2	18.9	16.7	16.8	20.3	24.2	24.6	24.1	24.2
		MCWB	16.9	16.4	15.3	13.0	9.5	7.3	6.9	8.9	10.8	13.3	15.5	16.4
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	19.3	18.9	18.6	16.8	13.7	10.7	10.2	13.1	15.1	17.1	18.3	19.2
		MCDB	24.3	24.2	23.4	21.3	18.8	17.2	16.9	20.8	22.8	23.6	24.0	25.3
	2%	WB	18.6	18.2	17.7	15.7	12.5	9.8	9.1	12.0	13.7	16.0	17.5	18.3
		MCDB	23.4	23.4	22.1	20.2	17.8	15.7	16.0	19.8	21.7	22.9	23.4	24.0
	5%	WB	18.0	17.7	17.0	14.9	11.6	9.1	8.4	11.0	12.8	15.2	16.9	17.7
		MCDB	22.8	22.5	21.0	18.9	17.0	15.0	15.1	18.3	21.2	21.8	22.7	23.1
	10%	WB	17.5	17.1	16.3	14.1	10.8	8.3	7.6	10.1	12.0	14.4	16.2	17.1
		MCDB	22.0	21.5	20.1	18.6	16.2	13.8	14.2	16.8	20.3	20.9	21.8	22.2
(35) Mean Daily Temperature Range	5% DB	MDBR	9.5	9.6	9.9	10.7	12.0	12.3	12.7	13.4	13.9	12.6	11.0	10.5
		MCDBR	12.2	12.2	12.6	12.8	13.4	13.9	14.4	15.0	15.7	15.6	13.6	13.1
		MCWBR	4.5	4.6	4.5	4.7	5.1	5.8	6.1	5.5	5.3	5.1	4.8	4.9
	5% WB	MCDBR	10.2	9.9	9.9	10.5	11.5	11.8	12.5	13.2	14.1	12.8	11.7	11.0
		MCWBR	4.3	4.3	4.4	4.4	5.0	5.5	5.6	5.2	5.8	5.4	4.9	5.0
(40) Clear-Sky Solar Irradiance	taub		0.383	0.373	0.367	0.326	0.288	0.271	0.271	0.328	0.374	0.418	0.381	0.388
	taud		2.314	2.343	2.346	2.428	2.512	2.530	2.514	2.348	2.249	2.168	2.303	2.293
	Ebn at Noon		959	953	928	922	921	918	932	905	906	899	956	957
	Edh at Noon		139	132	124	104	87	81	85	111	135	155	140	142
(44) All-Sky Solar Radiation	RadAvg		6.32	6.16	5.37	4.59	4.21	3.85	4.14	4.82	5.56	5.79	6.19	6.42
	RadStd		0.61	0.56	0.49	0.36	0.22	0.21	0.13	0.25	0.33	0.41	0.53	0.36

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (9 neighbors)		+0.66	+0.75	N/A	N/A	N/A	N/A	-181	+234	+87

Nomenclature: See separate page